

Cambridge International AS Level
Information Technology 9626 · Paper 1 Theory

Topic 8

Spreadsheets

11 questions from 9 question papers, May/June 2024 to May/June 2026.

Questions come first, in date order, then their mark schemes in the same order. Each question is labelled with the paper it came from. Click "Mark scheme" next to a question to jump to its answers, or use the bookmarks panel.

A question that covers more than one topic appears in each of those topics. Topics follow the 2025-2027 syllabus.

Questions

9626/13 May/June 2024 · Question 4

Mark scheme

4 Describe the terms absolute and relative cell replication, giving an example of each.

Absolute

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Relative

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9626/13 Oct/Nov 2024 · Question 6

Mark scheme

6 In a spreadsheet, a user has entered the following formula into a cell:

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=IF(NOT(OR(B6<5,B6>8)),"4th Quarter","")
```

Describe how the spreadsheet will calculate the result to be displayed.

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..... [3]

7 Test scores can be used to track a student’s progress over time. Describe **two** ways this can be done using a spreadsheet.

Way 1
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Way 2
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[4]

[6]

12 A spreadsheet is to be used to store students’ examination results.

Each examination is marked out of 60, with the lowest mark being 0.

Only whole marks are allowed.

Each mark is validated using this validation rule ≥ 0 and ≤ 60

The developer is trying to test the validation rule.

Complete the table for the **three** types of test data. For each mark the developer enters, suggest how the validation rule deals with the mark.

Type of test data	Example of mark entered	How the validation rule deals with the mark entered
Normal		
Abnormal		
Extreme		

[6]

- 2** The spreadsheet shows the prices of various items in a clothes shop. The discounted price for a jumper is calculated by using the formula in cell F12.

F12 : ✕ ✓ <i>f_x</i> =E12-(VLOOKUP(D12,\$D\$5:\$E\$8,2,FALSE)*E12)								
	A	B	C	D	E	F	G	H
1								
2								
3				Discount code				
4								
5				A	25%			
6				B	20%			
7				C	10%			
8				D	5%			
9								
10	Item				Price	Discounted price		
11								
12	Jumper			B	\$36.00	\$28.80		
13	Skirt			A	\$25.00	\$18.75		
14	Blouse			D	\$30.00	\$28.50		
15	Trousers			A	\$40.00	\$30.00		
16	Jeans			C	\$33.00	\$29.70		
17	Dress			B	\$50.00	\$40.00		
18	Trainers			A	\$55.00	\$41.25		
19	Shoes			D	\$60.00	\$57.00		
20								

- (a)** The formula in cell F12 is =E12-(VLOOKUP(D12,\$D\$5:\$E\$8,2,FALSE)*E12)

Describe how this formula works.

..... [5]

The formula in cell F12 is replicated to cells F13 to F19.

(b) In the formula, absolute cell referencing and relative cell referencing have **both** been used.

Explain why both types of cell referencing are needed.

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..... [4]

- 9 Eric has made a spreadsheet that he has shared with a number of friends. Some of his friends cannot open the spreadsheet.

Identify **three** possible reasons the spreadsheet will not open once received. For each reason, suggest a way Eric could have prevented the problem.

Reason 1
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Way 1
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Reason 2
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Way 2
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Reason 3
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Way 3
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[6]

..... [6

3 A teacher is creating a spreadsheet.

(a) Describe **two** ways in which the spreadsheet can validate data at the input stage.

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..... [2]

The spreadsheet will include both absolute cell referencing and relative cell referencing.

(b) Define the term absolute cell referencing.

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..... [1]

(c) Identify and explain **one** benefit of using relative cell referencing when creating a spreadsheet.

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..... [2]

(d) The teacher has used the following formula in a cell:

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=IF(C2="Dog",(IF(D2="Male","Block A","Block B")),(IF(D2="Male","Block C","Block D")))
```

Describe how the spreadsheet will produce the result to be displayed.

..... [6]

3 A student has created a spreadsheet to store data.

(a) Describe **two** features of spreadsheet software that can be used to make the process of entering data into the spreadsheet easier for the student.

feature 1

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feature 2

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[2]

The student needs to analyse the information she has entered.

(b) Describe **four** features of spreadsheet software that can be used to analyse the information that has been entered into the spreadsheet.

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[4]

- 8 A teacher has created a spreadsheet to store marks and grades achieved by students in an examination. Part of this spreadsheet is shown.

	A	B	C	D	E	F	G	H
1	Grade boundaries				Student grades			
2	Grade	Minimum Mark	Maximum Mark		Student name	Score	Grade	
3	A	44	50		Kyle			
4	B	40	43		Germain			
5	C	36	39		Jose			
6	U	0	35		Sandy			
7								

- (a) Describe the formatting that has been applied to the contents of cell B2.

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..... [2]

The teacher needs to test the spreadsheet. When testing a spreadsheet, it is important to use appropriate test data.

- (b) Describe **one** defining characteristic of each of:

- normal test data
- extreme test data
- abnormal test data.

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..... [3]

- (c) Complete the table below to identify **three** items of test data that can be used to test the mark range for a grade A.

test type	test data
normal	
extreme	
abnormal	

[3]

	A	B	C	D	E	F	G	H
1	Grade boundaries				Student grades			
2	Grade	Minimum Mark	Maximum Mark		Student name	Score	Grade	
3	A	44	50		Kyle			
4	B	40	43		Germain			
5	C	36	39		Jose			
6	U	0	35		Sandy			
7								

The teacher will use the following formula in cell G3 to show the grade achieved by Kyle.

=IF(F3>C3,"Check Score",(IF(F3>B3,A3,(IF(F3>B4,A4,(IF(F3>B5,A5,A6)))))))

- (d)** This formula has been tested and found not to work as intended.

Explain why this formula does not work as intended.

..... [5]

Mark schemes

Mark scheme · 9626/13 May/June 2024 · Question 4

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Question	Answer	Marks
4	Four from: Max two for each: <i>Absolute:</i> - Always copies the same cell reference without modification when replicated/a fixed reference to a cell (1) - Any example of an absolute value/formula/source cell not changing (1) <i>Relative:</i> - It copies the cell reference, but will increment the cell reference when replicated//cell reference changes as the formula location changes (1) - Any example of a relative value/formula/source cell changing (1)	4

Mark scheme · 9626/13 Oct/Nov 2024 · Question 6

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Question	Answer	Marks
6	THREE from: - It will check the contents of cell B6 (any awareness that outcome is based on contents of B6) (1) - If the contents are NOT less than 5 (1) OR NOT greater than 8 (1) (accept converse) ONLY award following if both above stated (NB note effect if answer above is converse) - The cell displays 4th Quarter (1) - if not/otherwise the cell displays a null string. (1) (Answer is dependant) - OR - If condition not met, a blank cell is displayed (1)	3

Question	Answer	Marks
7	FOUR marks available. TWO per matched answer. - Identify a suitable feature of a spreadsheet (1) - State how this feature can be used for the purpose (1) E.g.: Award first answer in each section only, unless no answer written in Way 2. - Manually store/save data about a student's scores/progression (1) to create a record of the student's performance. (1) - Automation/Formulas could be implemented in a spreadsheet (1) to calculate/predict averages of a given/specific time frame. (1) - Conditional formatting could be used in a spreadsheet (1) to highlight grades that meet a set criteria/condition. (1) - A graph/chart could be created from data (1st) - To represent/visualise a student's progression overtime in any given time/test. (1) - A complex/advanced spreadsheet with formulas that function across sheets could be created (1st) - To compare results/scores/progression/trends across different students/groups/classes/tests. (1) - Sort scores into date order (1) so progress over time can be seen (1)	4

Question	Answer	Marks
2	Max 4 marks: MUST be a comparison <i>Benefits:</i> • Easier to create a storage systems//multiple tables/worksheets (1) • Easier to calculate (1) • Easier to create complex formulae//carry out complex calculations (e.g. use functions (if state functions, this is equivalent to complex)) (1) • Repeated data can be easier to enter (1) • Charts are easier to produce (1) • Includes conditional formatting to (e.g.) highlight data easier (1) Max 4 marks: <i>Drawbacks:</i> • Information may exist across many sheets (1) • Large spreadsheets are more difficult to use/read (1) • Spreadsheets are best suited to single focus (e.g. per year) (1) • Formulae may need to be modified as spreadsheet grows (1) • Harder to create complex queries (1) • Harder to create a report (1) • Validation is easier in a database (1) • Repeated data exists in spreadsheet//databases are normalised (1) • Referential integrity reduces (e.g.) redundant data (1)	6

Question	Answer	Marks												
12	One mark per bullet point to a maximum of six marks. <i>Normal</i> • Data is accepted by the system • Any integer between 0 and 60(inclusive) <i>Abnormal</i> • Data outside of the range which will be rejected/not accepted by the system • Any data that is not an integer between 0 and 60 <i>Extreme</i> • Data that is at the boundaries of the acceptable range • 0 or 60 E.g. <table> <tr> <th>Type of test data</th><th>Example of mark entered</th><th>How the validation rule deals with the mark entered</th></tr> <tr> <td>Normal</td><td>41</td><td>Data is within the range data and will be accepted by the system</td></tr> <tr> <td>Abnormal</td><td>68/-2/pink</td><td>Data outside of the range which will be rejected/not accepted by the system</td></tr> <tr> <td>Extreme</td><td>0/60</td><td>Data that is at the boundaries of the acceptable range and will be accepted by the system</td></tr> </table>	Type of test data	Example of mark entered	How the validation rule deals with the mark entered	Normal	41	Data is within the range data and will be accepted by the system	Abnormal	68/-2/pink	Data outside of the range which will be rejected/not accepted by the system	Extreme	0/60	Data that is at the boundaries of the acceptable range and will be accepted by the system	6
Type of test data	Example of mark entered	How the validation rule deals with the mark entered												
Normal	41	Data is within the range data and will be accepted by the system												
Abnormal	68/-2/pink	Data outside of the range which will be rejected/not accepted by the system												
Extreme	0/60	Data that is at the boundaries of the acceptable range and will be accepted by the system												

Mark scheme · 9626/12 May/June 2025 · Question 2

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Question	Answer	Marks
2(a)	Max 4 from: VLOOKUP - It takes the B from cell D12 (1) - It looks for B in cells D5:E8 (1) - Returns/finds the corresponding value//20%//discount percentage (1st) - from column E/2nd column from the range (1) - It looks for an exact match//If it does not find an exact match it will return an error (1) Max 2 from: CALCULATION - It multiplies the percentage discount by the Price (1) - Subtracts discount from the original price. (1)	5
2(b)	Four marks from: Both are needed because some cell references need to remain the same (1) whilst others change (1). Absolute cell referencing always refers to the same cell therefore does not change (1st) - Valid example of a cell reference that must not change (1) Relative cell referencing changes depending on the position in the spreadsheet//may exemplify new positions (1st) - Valid example of a cell reference that must change. (1)	4

Mark scheme · 9626/12 May/June 2025 · Question 9

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Question	Answer	Marks
9	Max 6 from three matched pairs - Different software on Eric's computer than his friends' computers (1st) - E.g. use generic file type (1) - Saved in a different/wrong (file) format (1st) - E.g. use generic file format (1) - Anti-Virus does not allow file to open/File has a virus (1st) - E.g. Eric needs to virus check the file before sending it (1) - File is password protected/encrypted and Eric didn't share the password/decryption key (1st) - E.g. Eric needs to share the password (1). - Wrong access rights/permissions assigned to the file (1st) - E.g. grant access rights/permissions (1)	6

Question	Answer	Marks
5	One mark per bullet point to a maximum of six marks. Valid/Normal//Data that is from within the acceptable range (1st) • To test that the output is as expected/valid (1) • To test that it is accepted (1) Invalid/Erroneous//Data//Abnormal that is outside of the acceptable range (1st) • Data is of the wrong type/invalid (1) • To test that it is rejected (1) Boundary/Extreme//Data that is at the boundary (of the acceptable range) (1st) • To test that the boundaries are set correctly (1) • To test that the outcome is as expected – accepted and correct output (1)	6

Question	Answer	Marks	Guidance
3(a)	TWO marks available: • Use a drop-down menu which only allows selection from the list (1) • Accept only whole number e.g. between two extremes/greater than X (1) • Accept only decimal e.g. between two extremes/greater than X (1) • Accept only date e.g. between two extremes/equal to a set date (1) • Accept only time e.g. between two extremes/before a set time (1) • Restrict text length e.g. between two extremes/greater than X (1) • Set allowable value to be equal to that in another cell (1) • Use a (run on change) macro to check the value (1)	2	Do Not Award • 'Use validation tools' • Length check on its own, but allow description of length check
3(b)	ONE mark available: • A cell reference that will not change when the formula is dragged down//copied to another cell/replicated (1)	1	MUST have both parts

Question	Answer	Marks	Guidance
3(c)	TWO marks available for one complete explanation Identify benefit (1st) state why it is a benefit/state the beneficial implication (1) • Identification of benefit (1st): e.g. Fewer errors Reduced time to create the spreadsheet/formula • Explanation (1) e.g. Because formula only has to be typed once Because can enter formula in one cell and can (e.g.) drag down Because cell references automatically change as copied/dragged/replicated	2	Do Not Award • Makes data more accurate, however, if this is further explained – such as fewer errors, you may award fewer errors • Enter numbers and drag down – has to be formulas
3(d)	SIX marks available: • Checks value from cell C2 (1) • If the contents are Dog (1) • The first IF statement is run (1) • It will check the contents of cell D2 (1) • If the contents are Male (1st) It will display “Block A” (1) If not, it will display “Block B” (1) • If C2 not Dog (1) • It will check contents of cell D2 AND if the contents are Male (1st) It will display Block C otherwise it will display Block D (1)	6	

Question	Answer	Marks	Guidance
3(a)	TWO marks available: • (Data) import from a different source/spreadsheet/database (1) • User forms which provide input boxes/radio buttons (1) • Macros to run input boxes (1) • Drop down lists from which can select an answer (1) • Input boxes linked to specific cells (1) • Copy and paste from another source/spreadsheet/database (1) • Autofill populate cells based on a set pattern/repeated data (1) • Fill series populate cells based on a set pattern (1) • Radio buttons to select options (1)	2	

Question	Answer	Marks	Guidance
3(b)	FOUR from: - Spreadsheet software allows what-if analysis to be carried out (award any mention of what if analysis) (1) - Can sort data into a rank order (concept of ordering) (1) Can use macros which can create routines /subroutines (1) - Macros/subroutines can carry out fixed routines//e.g. goal seek (1) - Conditional formatting can show/highlight data ONE FROM that matches//is greater than//less than criteria (1) based on rules/conditions (may exemplify) (1) - Carries out calculations such as subtraction (accept any example of a calculation) (1) - Formulae can be created for calculations (1) - Functions can perform set tasks OR (one mark in total) Uses Functions such as OR (one mark in total) Identifies a function by name and states what it does (1) - Charts/graphs to present values//allow for comparison//visualise data (1) - Pivot Tables to analyse large quantities of data in a visual form (1) - Filtering based on set criteria (may exemplify e.g. for a certain class) (1)	4	Award first four attempts only Do Not Award: Features which are NOT directly to do with analysis

Question	Answer	Marks	Guidance
8(a)	TWO from: - In bold (1) Centrally aligned//centred horizontally/vertically (1)	2	Do Not Accept Border – not a content
8(b)	THREE marks available: Normal test data (max ONE) - Data that should produce expected results (1) - Data that is within acceptable boundaries//correct range (1) - Should be accepted by the cells (1) - Sensible data (1) Extreme test data (max ONE) - Data that is at the boundary (1) - At the limit of acceptability (1) Abnormal test data (max ONE) - Falls outside acceptable limit/range (1) - Data that is invalid (1)	3	
8(c)	THREE marks available, one per type of test: Normal - Any single number between 44 and 50 (inclusive) (1) Extreme 44//50 (ONE ONLY – NOT BOTH) (1) Abnormal - Any other single number beyond range 44 - 50 (1) - Any data other than numeric (1)	3	Accept real values between 44 and 50 Answer must be actual data (e.g. 50) and not description of data (e.g. 'a value between 44 and 50'). If descriptions given, do not accept this as non-numeric for abnormal test data

Question	Answer	Marks	Guidance
8(d)	FIVE from: =IF(F3>C3,"Check Score",(IF(F3>B3,A3,(IF(F3>B4,A4,(IF(F3>B5,A5,A6))))))) - The formula will not produce the correct grade for all scores (1) at the (lower) boundary of each grade (1) it will produce the grade below (1) Wrong inequality symbol marks (route 1) - The formula only uses an inequality symbol (1) and should have used an inequality (accept equivalent statement, such as > has been used) AND an equals statement (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) Wrong number marks (route 2) - The wrong value has been used as the lower boundary (1) - The formula uses a greater than inequality (1) the value from the top of the lower boundary should be used (1) - Therefore only marks above the boundary are awarded the grade (1) - An (e.g.) A grade piece of work, gets a B grade. (1) One mark may be awarded for: - One mark for showing how the formula may be improved:	5	Marks may be awarded from route 1 OR route 2 only. Where candidate mixes both explanations, mark to the candidate's best advantage.

Question	Answer	Marks	Guidance
8(d)	e.g. • IF(F3 > B3 – 1 • IF(F2 = B3 etc. IF(F2 > 43		